

Evolution of Visual Representation Learning and Its Applications in Remote Sensing

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Abstract—Visual Representation Learning (VRL) is essential for transforming raw satellite imagery into actionable geographical intelligence. While deep learning has revolutionized remote sensing (RS), the “label scarcity” challenge has shifted the field from supervised backbones to self-supervised learning (SSL) and large-scale foundation models. This paper reviews the evolution of VRL in RS, categorizing methodologies into supervised, reconstruction-based, and similarity-based paradigms. We specifically analyze how general computer vision techniques are adapted to handle the unique spectral, spatial, and temporal dimensions of Earth observation data. Finally, we discuss emerging trends, including cross-modal alignment and spatio-temporal dynamics, providing a roadmap for developing universal foundation models for planetary-scale monitoring.

Index Terms—Visual Representation Learning, Remote Sensing, Self-Supervised Learning, Foundation Models, Transfer Learning

I. INTRODUCTION

Visual Representation Learning (VRL) serves as the cornerstone of modern computer vision, aiming to map high-dimensional raw pixels into compact, semantically meaningful feature spaces [1]. While deep learning has revolutionized image interpretation, the performance of downstream tasks—such as classification, detection, and segmentation—is fundamentally constrained by the quality of the learned representations. In the context of Remote Sensing (RS), the drive toward advanced VRL is propelled by a unique paradox: the availability of massive volumes of satellite imagery contrasted with a critical shortage of expert-annotated labels [2]. Unlike natural images, RS data encompasses diverse modalities (multispectral, SAR), varying spatial resolutions, and a distinct nadir (top-down) perspective, making effective representation learning essential for bridging the gap between raw data and actionable geographical intelligence.

Historically, the RS community has relied on transfer learning from models pre-trained on large-scale natural image datasets like ImageNet. While low-level visual primitives are often shared across domains, the “domain gap” between terrestrial photographs and orbital observations limits the efficacy of generic features. This has sparked an evolution in VRL paradigms within the RS field. The trajectory has shifted from traditional supervised backbones to self-supervised learning (SSL) strategies that exploit the inherent structure of unlabeled

data [3], [4], and most recently, toward the development of Large-scale Foundation Models. These models, such as the DINO series [5], [6] and domain-specific variants like AlphaEarth [7], demonstrate unprecedented generalization capabilities by learning from vast, heterogeneous datasets.

To provide a clear roadmap for researchers navigating this rapidly evolving landscape, this review offers a structured analysis of VRL with an emphasis on its adaptation to the remote sensing domain. The main contributions of this work are summarized as follows:

- **Systematic Categorization of VRL Paradigms:** We provide a comprehensive taxonomy of representation learning, ranging from supervised architectures to the latest self-supervised and foundation model frameworks.
- **Domain-Specific Adaptation Analysis:** We analyze how general computer vision methodologies are tailored to handle the unique spectral, spatial, and temporal characteristics of remote sensing imagery.
- **Future Outlook on Emerging Trends:** We identify critical research frontiers, including cross-modal representation learning, spatio-temporal dynamics, and the integration of physical priors into deep learning models.

Fig. 1 illustrates the overall structure of this survey and highlights key papers discussed in each section.

II. CONCEPTUAL FRAMEWORK OF VISUAL REPRESENTATION

A. Definition of Representation

At its core, visual representation learning aims to find a transformation function f that maps a raw input image x from the high-dimensional pixel space $\mathcal{X}$ to a representation/feature in a lower-dimensional feature space $\mathcal{Z}$:

$$f : \mathcal{X} \rightarrow \mathcal{Z}, \quad x \mapsto z = f(x) \quad (1)$$

Then the learned representation z can be utilized for various downstream tasks, such as image classification, object detection, and semantic segmentation. The quality of the representation is often evaluated based on its ability to facilitate these tasks effectively [3], [8].

B. From Handcrafted to Learned Representations

Historically, visual representations were synonymous with handcrafted descriptors. Classic algorithms such as Scale-Invariant Feature Transform (SIFT) [9] and Histogram of Oriented Gradients (HOG) [10] relied on fixed, human-engineered mathematical operations to extract local primitives like edges and corners. Although these methods have been found useful in the field of remote sensing, their “shallow” nature limited their ability to represent complex, high-level semantic concepts.

The advent of Deep Neural Network (DNNs) shifted the paradigm from “feature engineering” to “feature learning.” In a deep architecture, the function f is composed of multiple non-linear layers:

$$f(x) = f_L \circ f_{L-1} \circ \dots \circ f_1(x). \quad (2)$$

These deep learning methods excel traditional handcrafted methods in multiple computer vision tasks like classification [11], [12], [13], object detection [14], [15] and segmentation [16]. Former studies [17], [18] show that this structure enables the model to learn hierarchical representations, where lower layers capture basic visual patterns, and higher layers encode more abstract concepts.

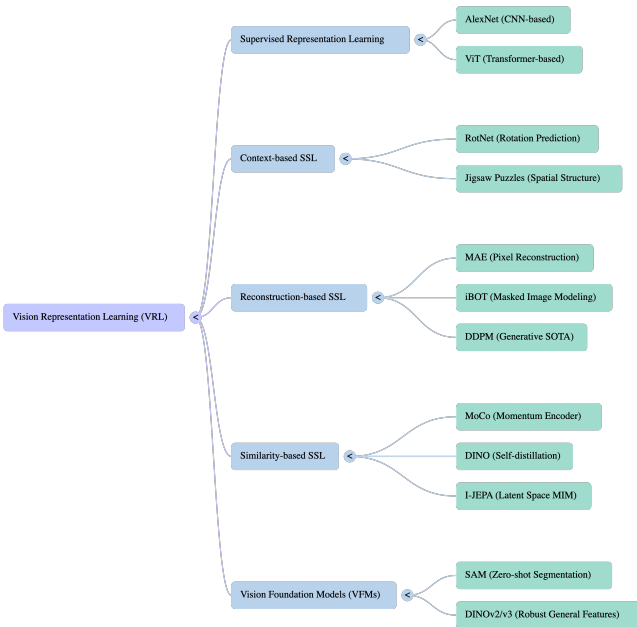


Fig. 1. **General Structure** of this survey and the **Important Papers** covered in each section.

C. Utility in Downstream Applications

The power of representation learning lies in its utility for downstream tasks, which is also known as transferability or generalizability [19]. A superior representation is one that makes the subsequent task (e.g., classification) “linearly separable”, meaning that the learning algorithms have captured the underlying information that is “enough” for different downstream tasks, as illustrated in Fig. 2. Former studies [20], [21] demonstrate that features extracted from pre-trained models can significantly boost performance on various tasks, even

with limited labeled data. This is particularly beneficial in remote sensing, where annotated datasets are often scarce and expensive to obtain.

The transferability of learned representations has inspired the “pre-training and fine-tuning” paradigm, where models are first trained on large-scale datasets (e.g., ImageNet [22]) and then fine-tuned on specific tasks with smaller datasets. This approach has become a standard practice in both computer vision and remote sensing communities, enabling the development of robust models that can adapt to diverse applications with minimal additional training.

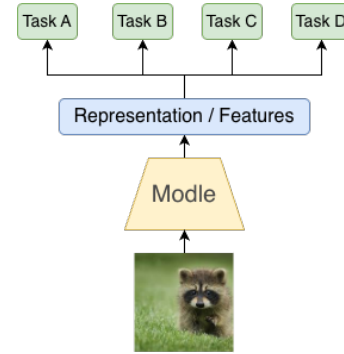


Fig. 2. **Transferability of learned representation.** A model can be trained on source task(e.g. ImageNet classification), and then the output representation can be transferred to various downstream tasks(e.g. remote sensing image classification, object detection, and semantic segmentation).

D. The Convergence of VRL and Remote Sensing

Remote sensing benefits from VRL in two main aspects. First, VRL methods can be directly applied to train models on large-scale remote sensing datasets, extracting more powerful and generalizable features compared to traditional supervised learning approaches. Second, large pre-trained models on natural image datasets can be fine-tuned or adapted to remote sensing tasks, leveraging the learned representations to overcome the challenges posed by limited labeled data in this domain. By unifying diverse spectral, spatial, and temporal information into a compact latent space, these VRL paradigms bridge the gap between raw Earth observation data and high-level semantic understanding, providing the foundational backbones for the specific learning methodologies discussed in the following sections.

III. SUPERVISED REPRESENTATION LEARNING

Early visual representation learning predominantly relied on supervised learning paradigms, where models were trained on large labeled datasets to learn discriminative features. This section explores the architectural evolution from CNNs to ViTs and the transition toward large-scale supervised foundation models.

A. Architectural Foundations: CNNs and ViTs

Convolutional Neural Networks were proposed back in 1980s [23], and have been the backbone of supervised representation learning since the success of AlexNet [11]. Several following work [12], [13] further refined CNN architectures,

introducing deeper networks and residual connections that significantly improved feature extraction capabilities. CNNs leverage local connectivity and weight sharing to capture spatial hierarchies in images, making them particularly effective for visual tasks.

Vision Transformers (ViTs) [24] discard the convolution operation in favor of transformer, which was first introduced in natural language processing. By treating an image as a sequence of patches, ViTs capture global dependencies through Self-Attention operation. The operation is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

where Q , K , and V represent the query, key, and value matrices derived from the input embeddings, and d_k is the dimensionality of the key vectors.

Following work [25], [26], [27] enhanced ViTs with hierarchical structures and efficient attention mechanisms, enabling them to capture long-range dependencies and global context more effectively than CNNs. ViTs have demonstrated superior performance on various benchmarks, especially when trained on large-scale datasets.

Remote Sensing. Supervised VRL is typically implemented in RS through two paths. The first is to train models on large-scale RS datasets [28], [29], [30] with abundant annotations. The second is to fine-tune pre-trained models on natural image datasets like ImageNet [22] for specific RS tasks. Most of the early RS work [30], [31] adopted CNN-based architectures for tasks such as land cover classification and object detection. U-net [32], a CNN-based structure first introduced to biomedical image segmentation, has made it possible to process high-resolution remote sensing images [33], [34], [35]. More recent studies [36], [37], [38] have explored ViT-based models, demonstrating their effectiveness in capturing the complex spatial patterns present in RS imagery. Multi-spectral data shares similar characteristics with natural images, making it feasible to directly reuse general computer vision models, both their architectures and pre-trained weights, to remote sensing tasks with minor modifications. However, the uniqueness of SAR and hyperspectral data often requires training from scratch. Several studies [39], [40], [41], [42], [43] have designed specialized supervised architectures (both CNN and ViT) to better capture the distinct features of these modalities.

B. Large-scale Labeled Dataset and SAM

The success of LLMs in natural language processing has inspired the development of large-scale supervised foundation models in computer vision. These models are trained on massive datasets with billions of images and labels, enabling them to learn rich and diverse representations that generalize well across various tasks. SAM (Segment Anything Model) [16], [44] proposed a pipeline to create a large-scale segmentation dataset and trained a ViT-based model that can generalize to various segmentation tasks with minimal fine-tuning, which showed “zero-shot” capabilities, meaning great transferability to unseen tasks without additional training.

Remote Sensing. In RS domain, the main problem is that there is no large-scale labeled dataset comparable to SAM’s dataset. Several studies [45], [46] have attempted to add lightweight adapters to the original SAM model and fine-tune it on RS segmentation datasets, achieving promising results. Other studies [47], [48], [49] have explored LoRA and prompt engineering techniques to adapt SAM for RS applications. Beyond segmentation, some work has investigated the use of SAM’s backbone for other tasks like object detection [50], [51] and moving object tracking [52], demonstrating the versatility of large-scale supervised foundation models in remote sensing.

Though supervised representation learning has achieved significant success, its reliance on large labeled datasets poses challenges in RS, where annotations are often scarce and expensive to obtain. This limitation has driven the exploration of self-supervised learning paradigms, which leverage unlabeled data to learn effective representations, as discussed in the next section.

IV. SELF-SUPERVISED REPRESENTATION LEARNING

Self-supervised learning (SSL) aims to learn representations by solving “pretext tasks” where the supervision is automatically generated from the data itself. In the RS domain, where unlabelled data is abundant but expert annotation is scarce, SSL has become the primary driver for building robust, domain-specific backbones.

We coarsely categorize SSL methods into three groups: **Context-based**, **Reconstruction-based** and **Similarity-based**, inspired by previous work [3], [4]. The overview of self-supervised VRL methods is illustrated in Fig. 3.

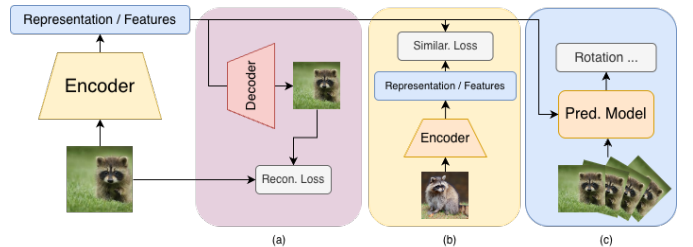


Fig. 3. **Self-Supervised VRL.** (a) Reconstruction-based. (b) Similarity-based. (c) Context-based.

A. Context-based Methods

Context-based methods rely on predicting certain properties of the input data, such as spatial location, rotation angle, or relative position of patches. By solving these pretext tasks, models learn to capture the underlying structure and semantics of the images.

Rotation. RotNet [53] is a pioneering work that trains models to predict the rotation angle of an image as a 4-category classification ($0^\circ, 90^\circ, 270^\circ, 360^\circ$) problem. The model learns to recognize object orientations, which helps in learning meaningful features. Following work [54], [55] refined rotation-based tasks by introducing more granular angle predictions or combining rotation with other pretext tasks.

Relative Position Prediction. Relative position prediction was proposed in [56], where the model learns to predict

the relative spatial arrangement of image patches. This task encourages the model to understand the spatial context and relationships between different parts of an image. In [57], the authors improved upon this idea by incorporating more complex sampling strategies and augmentations to enhance the learning process. And in [58], the same idea was extended to video data, where the model predicts the temporal order of video frames, capturing motion dynamics and temporal dependencies.

Jigsaw Puzzle. The Jigsaw puzzle task was introduced in [59], where an image is divided into several patches that are then shuffled. The model is trained to predict the correct arrangement of these patches. This task encourages the model to understand spatial relationships and object structures within the image. In [60], the authors extended this idea to temporal jigsaw puzzles for video data, demonstrating its effectiveness in learning spatio-temporal features.

Remote Sensing. Context-based methods have been widely adopted in RS for learning spatial and temporal features. Though RS images are often rotation-invariant, previous studies [61], [62] have shown that with proper data augmentation and training strategies, rotation-based SSL can still yield effective representations. In [63], [64], relative position prediction has been successfully applied to RS images, both in single images and time-series data, capturing spatial dependencies crucial for downstream tasks. Jigsaw puzzle tasks have also been explored in [65], in which the model is pre-trained with jigsaw puzzles on SAR images and then fine-tuned to do classification.

B. Reconstruction-based Methods

Reconstruction-based methods train models to reconstruct the original input from a corrupted or transformed version. This kind of pretext task dates back to denoising autoencoders [66], where the model learns to remove noise from images, thereby capturing essential features. Most of the reconstruction-based models rely on an encoder-decoder architecture, where the encoder maps the corrupted input to a latent representation, and the decoder reconstructs the original input from this representation. Reconstruction-based methods are particularly effective in learning low-level features and structures, and have been proved useful in both natural images and remote sensing images.

MIM (Masked Image Modeling) . Inspired by the success of BERT [67] in NLP, Masked Image Modeling (MIM) has been proposed for visual representation learning. In [68], the authors introduced BEiT, which masks random patches of an image and trains a ViT model to predict the missing patches using a discrete visual vocabulary. Following work [69], [70] further refined MIM techniques by exploring different masking strategies, prediction targets, and architectures. Especially, MAE [69] discarded the use of a visual vocabulary and directly reconstructed pixel values from masked patches, demonstrating the effectiveness of simple reconstruction objectives. VideoMAE v1-v2 [71], [72] extended MIM to video data, where the model learns to reconstruct masked regions across video frames, capturing both spatial and temporal features. SiamMAE [73] combined MIM with a siamese network

architecture, enhancing the model’s ability to learn invariant features. In [74], the authors proposed MultiMAE, which jointly learns to reconstruct multiple modalities (e.g., RGB, depth, segmentation masks) from masked inputs, demonstrating the versatility of MIM in multi-modal learning, which is particularly relevant for remote sensing applications involving multispectral and SAR data.

Generative Models. Generative models, such as VAEs [75], GANs [76] and Diffusion Models [77], though first proposed for data generation, have also been employed for representation learning. VAEs learn to encode input data into a latent space and then decode it back to reconstruct the original input, capturing essential features in the latent representation. GANs consist of a generator and a discriminator that compete against each other, leading to the learning of rich representations in the generator’s latent space. Diffusion models learn to generate data by reversing a gradual noising process, and their intermediate representations have been found useful for various tasks. Previous studies [78], [79], [80] have explored the use of generative models for monocular depth estimation, lighting estimation, texture extraction and other vision tasks.

Remote Sensing. MIM has been widely used in RS. In [81], researchers proposed RINGMo, a MIM-based model pre-trained on large-scale multispectral RS images, demonstrating superior performance on various downstream tasks such as land cover classification and object detection. Following studies [82], [83] explored methods to deal with multi-scale remote sensing images. Several work [84], [85], [86] have extended MIM to temporal and multi-modal RS data (including SAR images), capturing spatio-temporal dynamics and cross-modal relationships. Previous studies also showed that both the weights and architectures of MAE pre-trained on natural images can be reused to RS tasks. Generative models have also been explored in RS. In [87] and [88], researchers employed GANs and SOTA conditional diffusion models to transfer SAR images to optical images, achieving impressive results in image translation and enhancement tasks. Other studies [89], [90], [91] have utilized diffusion models for super-resolution and building extraction from RS images, demonstrating the versatility of generative models in remote sensing applications.

C. Similarity-based Methods

Similarity-based methods, also known as contrastive learning, aim to learn representations by bringing similar samples closer in the feature space while pushing dissimilar samples apart. This is typically achieved by defining positive and negative pairs of samples and optimizing a contrastive loss function. This methodology has gained significant attention due to its effectiveness in learning discriminative features without explicit labels, and become the most popular SSL to train foundation models in recent years.

The loss function commonly used in similarity-based methods is the InfoNCE loss [92], defined as:

$$L_{\text{contrast}} = -\log \left(\frac{\exp\left(\frac{\text{sim}(z_i, z_j)}{\tau}\right)}{\sum_{k=1}^{2N} 1_{[i \neq k]} \exp\left(\frac{\text{sim}(z_i, z_k)}{\tau}\right)} \right) \quad (4)$$

where $\text{sim}(z_i, z_j)$ measures the similarity between representations z_i and z_j (e.g., cosine similarity), τ is a temperature parameter, and N is the total number of samples in the batch.

Negative Samples. In negative sample-based methods, the model learns to distinguish between positive pairs (different augmentations of the same image) and negative pairs (augmentations from different images). MoCo [93] introduced a momentum encoder to maintain a dynamic dictionary of negative samples, enabling the use of smaller batch sizes in a larger training dataset. SimCLR [94] proposed a simpler framework that relies on large batch sizes to provide sufficient negative samples, demonstrating that with proper data augmentation and normalization, contrastive learning can achieve competitive performance. Following MoCo v2 and MoCo v3 [95], [96] further refined the MoCo framework by incorporating stronger data augmentations and exploring different architectures (Swin-Transformer [38]), achieving state-of-the-art results on various benchmarks.

Self-Distillation. Following the success of knowledge distillation [97] in supervised learning, self-distillation methods have been proposed for SSL, where a student model learns from a teacher model that is updated based on the student's parameters. These self-distillation methods often discard negative samples, which enables training on large-scale datasets. BYOL [98] introduced a framework where the student and teacher networks are trained to predict each other's representations without using negative samples. SimSiam [99] further simplified this approach by removing the momentum encoder and demonstrating that stop-gradient operations are sufficient to prevent collapse. DINO [100] applied self-distillation to ViTs, showing that it can learn meaningful representations that capture semantic information. iBOT [101] extended DINO by incorporating MIM objectives, enabling the model to learn both global and local features effectively. Following work [5], [6] further scaled up the DINO framework, training large ViT-based models with multiple regularization techniques to enhance representation quality. Unlike methods mentioned above, I-JEPA [102] proposed a joint-embedding architecture where the model learns to predict the representation of masked patches from unmasked context in a latent space, aiming to focus on high-level semantics rather than low-level pixel reconstruction.

Remote Sensing. Similarity-based methods have been extensively applied in RS for learning robust and transferable representations. Unlike natural images where "positive views" are created via random cropping, RS offers unique opportunities for pair construction due to its multi-temporal and multi-modal nature. Several studies [4], [37], [103] have leveraged multi-temporal and multi-modal RS images as image pairs, capturing temporal dynamics and seasonal variations. Other studies [104], [105]. Since DINO series [5], [6] have shown great transferability, some recent work [106], [107] have directly fine-tuned these models on RS tasks, achieving excellent performance.

V. CONCLUSION AND FUTURE PERSPECTIVES

The evolution of Visual Representation Learning in remote sensing marks a shift from capturing static textures to distilling the complex, multi-dimensional semantics of the Earth's surface. From the early days of handcrafted descriptors to the current dominance of self-supervised Vision Transformers, the primary objective has remained consistent: to build a robust bridge between raw pixels and geographical intelligence. As we move beyond the era of task-specific models trained on isolated datasets, the field is converging toward a unified paradigm of Universal Representation Learning. This shift is most prominently evidenced by the rise of Vision Foundation Models (VFMs) and Vision-Language Foundation Models (VLFMs), which serve as the definitive frontier for next-generation remote sensing.

Looking forward, the development of these foundation models is not merely about scaling data, but about enriching the representation with broader contexts and constraints. First, the expansion toward cross-modal and multi-sensor foundation models [81], [82] will enable a holistic understanding of the Earth by aligning optical, SAR, and LiDAR data into a single latent space, thereby overcoming the limitations of any single sensor. Second, the integration of Vision-Language alignment [108] will facilitate "open-vocabulary" interpretation, allowing for human-centric reasoning and natural language interaction with petabytes of satellite imagery.

Beyond scaling, the future of VRL must also address the inherent physical and temporal properties of our planet. A critical direction is the transition to spatio-temporal representation learning, where models learn the "dynamic rhythm" of the Earth—capturing phenological cycles and urban evolution rather than static snapshots. Furthermore, the incorporation of physics-based priors and process understanding into the learning objective will ensure that the learned features respect geophysical laws, bridging the gap between data-driven AI and traditional Earth system science. Finally, as we look toward real-time global monitoring, the dual advancement of lightweight on-board VRL for satellite edge computing [109] and causal reasoning capabilities [7], [110] to transform remote sensing from a reactive mapping tool into a proactive, intelligent system for global decision-making.

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